

Relational Dynamics and the Physics of Consciousness: A Bottom-Up Formalism

Introduction

The scientific and philosophical pursuit of a physical theory of consciousness has historically fractured along profound interdisciplinary fault lines. On one extreme, computational functionalism abstracts the mind into substrate-independent algorithms, divorcing subjective experience from its physical embodiment. On the other extreme, neurobiological reductionism catalogs cellular minutiae without articulating the macroscopic physical laws that bind disparate neuronal firings into a unified conscious percept. This analysis advances a complete, purely relational, bottom-up physical formalism of consciousness. Rather than positing consciousness as a computational output, an epiphenomenal illusion, or a fundamental substance like charge or mass, this framework establishes subjective experience as an emergent, thermodynamic, and topological property of open, non-equilibrium physical systems.

By grounding the architecture of the mind in the most fundamental physical primitives—beginning with the Poincaré group and Noether's theorem—this framework rigorously traces the physical evolution of sentience. It traverses the spontaneous symmetry breaking (SSB) of quantum fields, the emergence of non-Abelian gauge theories, and the stabilization of macroscopic memory domains through geometric frustration. Furthermore, it explicitly links these topological structures to the inexorable thermodynamic constraints of information processing, specifically Landauer erasure, which enforces a mandatory coupling between logical state changes and physical heat dissipation. The formalism scales into the macroscopic realm through boundary back-action, mathematically formalized via Markov blankets, Bayesian mechanics, and the generalized synchronization of macroscopic neural assemblies.

Ultimately, this thermodynamic and topological architecture serves not merely as an explanatory model, but as a rigid physical diagnostic tool. It systematically demonstrates why rigid von Neumann architectures, such as Graphics Processing Units (GPUs) executing Large Language Models (LLMs), are intrinsically disqualified from achieving consciousness due to their broken thermodynamic feedback loops. In contrast, mortal computation—where hardware and software are deeply and inextricably coupled—emerges as the necessary and sufficient physical substrate for sentient systems.

Foundational Symmetries and Physical Primitives

The Poincaré Group and the Relational Nature of Spacetime

Any bottom-up physical formalism must begin at the foundational symmetries that govern space, time, and matter. In modern physics, the baseline reality of the universe is not described by static objects, but by relational fields constrained by symmetries. The most fundamental of these is the Poincaré group, denoted mathematically as $ISO(3,1)$, which represents the

continuous symmetries of Minkowski spacetime. The Poincaré group comprises ten generators: four for translations in space and time, three for spatial rotations, and three for Lorentz boosts. In standard Quantum Field Theory (QFT), the invariance of a closed physical system under the Poincaré group ensures the conservation of fundamental quantities. The evolution of a physical system is described by an action functional $S = \int \mathcal{L} \, d^4x$, where $\mathcal{L}$ is the Lagrangian density. Through Noether's theorem, every continuous symmetry of the action yields a conserved current. For instance, translational invariance in spacetime generates the conserved stress-energy tensor $T^{\mu\nu}$, satisfying the continuity equation $\partial_\mu T^{\mu\nu} = 0$. This gives rise to conserved energy and momentum. Similarly, invariance under global phase transformations (gauge symmetries) generates conserved charge currents J^μ . However, conscious biological systems are not closed; they are fundamentally open, dissipative, non-equilibrium thermodynamic systems. They exist in continuous, unavoidable interaction with their external environments. For such open systems, the strict global symmetries of the Poincaré group are explicitly and spontaneously broken at the local scale. The stress-energy tensor is no longer strictly conserved within the arbitrary boundaries of the brain, as there is a continuous flux of energy and entropy across the neurobiological boundary. This profound relational openness necessitates a fundamental shift from the conservative, closed-system Hamiltonian dynamics of classical mechanics to the physics of dissipation, open systems, and spontaneous symmetry breaking.

From Global Symmetries to Local Gauge Invariance

When a global symmetry is subjected to local perturbations—such as sensory inputs impinging upon a biological system—the symmetry is threatened. To preserve the invariance of the Lagrangian under local transformations, physics introduces gauge fields. A gauge theory predicts how physical fields interact with matter to maintain symmetry despite local deformations.

If we consider the brain as a physical system attempting to maintain a stable internal state (homeostasis) in the face of continuous environmental perturbations, the internal dynamics of the brain act analogously to a gauge field. When local forces act upon the system, generating local transformations that break global symmetries, the system must deploy a counteracting force—the gauge field—to "repair" the broken symmetry and preserve the system's structural integrity. This relational dynamic between local perturbation and gauge-field compensation forms the microscopic foundation of what, at a macroscopic scale, is recognized as perception and active inference.

Symmetry Type	Transformation	Conserved Quantity / Field Role	Implication for Conscious Systems
Spacetime (Poincaré)	Translations & Lorentz Boosts	Energy-Momentum ($T^{\mu\nu}$)	Establishes the energetic constraints of the physical substrate.
Global Phase	U(1) or SU(N) Global	Conserved Charges (J^μ)	Defines the invariant identity of the closed universe.
Local Gauge	Spatially/Temporally Varying	Gauge Fields (A_μ)	Drives internal models to compensate for sensory perturbations (active inference).

The Dissipative Quantum Model of the Brain

Infinite Degrees of Freedom and Unitarily Inequivalent Vacuums

To bridge the immense explanatory gap between microscopic physical primitives and macroscopic cognitive phenomena, this analysis leverages the dissipative quantum model of the brain, pioneered by Ricciardi and Umezawa and significantly expanded by Giuseppe Vitiello. The foundational observation of this model is that the unavoidable dissipative character of brain functioning is not a thermodynamic nuisance or a source of noise, but rather the essential root of the brain's vast memory capacity and its conscious dynamics.

In standard quantum mechanics, which deals with a finite number of degrees of freedom, the Stone-von Neumann theorem guarantees that all representations of the canonical commutation relations are unitarily equivalent. This implies a single, unique vacuum state $|0\rangle$. However, in macroscopic systems characterized by infinitely many degrees of freedom—such as the densely interconnected fields of the mammalian brain—the Stone-von Neumann theorem breaks down. This breakdown permits the existence of a continuous spectrum of distinct, unitarily inequivalent vacuum states.

When a continuous symmetry of the system is spontaneously broken—meaning the underlying Lagrangian remains invariant but the physical ground state does not—the system undergoes a phase transition. According to Goldstone's theorem, this spontaneous symmetry breaking (SSB) requires the emergence of massless bosons, known as Nambu-Goldstone (NG) bosons. These bosons condense into the vacuum state, establishing a macroscopic, long-range order. In the context of the brain, these condensed NG bosons manifest as collective modes propagating across the neural substrate, physically encoding memories as robust macroscopic phase states rather than localized microscopic engrams.

The Thermodynamics of the "Double"

Because the brain is continuously dissipating heat and exchanging information with its environment, it cannot be modeled by a standard, closed-system Hamiltonian. The dissipative QFT framework solves this by formally modeling the environment as a time-reversed copy of the system, termed the "Double".

The total Hamiltonian H_{total} is constructed as the sum of the system's Hamiltonian H_{sys} , the environment's Hamiltonian H_{env} , and an interaction term H_{int} . Because the brain is an open system, it continuously exchanges energy with the Double. To maintain the canonical formalism of QFT, the state space of the brain and its environment is intertwined using Bogoliubov transformations. This leads to the generation of a generalized coherent state where the brain and the Double are perfectly entangled. Consciousness, within this physical paradigm, is postulated to arise directly from the permanent, dynamic "dialogue" between the brain and its environment. The continuous dissipation of energy drives the system through a trajectory of unitarily inequivalent vacuum states, each representing a distinct memory or cognitive phase.

Non-Abelian Self-Interaction and Topological Defect Memory

Classical Dissipative SU(2) Gauge Theory

While early QFT models of the brain relied heavily on simple Abelian symmetries (like $U(1)$), the profound complexity of cognitive state spaces requires a non-Abelian extension. Brain processes and the non-linear dynamics of consciousness can be mathematically formalized by a classical, dissipative, non-Abelian gauge theory, specifically an $SU(2)$ gauge theory. In an $SU(2)$ gauge framework, the associated gauge fields exhibit inherent self-interaction, a hallmark of non-Abelian Yang-Mills theories. Unlike photons in quantum electrodynamics, which do not carry charge and thus do not interact with one another, the gauge bosons of an $SU(2)$ theory carry the "charge" of the field itself, allowing them to couple to one another. This self-interaction introduces profound, complex nonlinearities into the system's dynamics, analogous to the highly recurrent, reentrant signaling pathways observed in the cerebral cortex. To accommodate the localized, non-relativistic nature of biological substrates, a specific ansatz for the gauge field is introduced, which explicitly breaks Lorentz invariance. Within this ansatz, a mathematical contraction mapping acts as the formal equivalent of thermodynamic dissipation. As the biological system evolves toward the attractive fixed point of this contraction mapping—representing the limit of maximal dissipation—the classical gauge fields reduce, up to constant proportionality factors, to the Pauli quantum gates for one-qubit states. This remarkable mathematical convergence implies that a classical, non-Abelian dissipative gauge theory possesses a hidden quantum nature, which is revealed precisely through its dissipation at temperatures approximating absolute zero within the quantum vacuum, where the coherence of quantum states is rigorously preserved. Consequently, the macroscopic classical dissipative system functions as a "quantum metalanguage" relative to the underlying microscopic logic of the neural substrate, physically instantiating the analog, non-algorithmic aspects of the conscious mind.

Memory Domains and Topological Robustness

The spontaneous breakdown of symmetry in this gauge-theoretic framework generates observable ordered patterns across the neural tissue. An order parameter emerges, acting as a macroscopic variable that specifies the physical phase of the vacuum by dictating the density of condensed NG bosons.

These bosons propagate over the entire system as collective modes, dynamically organizing memories in both space and time. Rather than being stored in localized single neurons, memory exists as a system-wide phase state—a physical memory domain. The time-dependence of the frequencies of the dipole wave quanta leads to the spatial localization of these domains and tightly dictates their lifetime.

From a mathematical perspective, these memory domains are equivalent to topological defects. In gauge theories and condensed matter physics, topological defects (such as domain walls, vortices, or hopfions) emerge when a symmetry is broken, locking the system into a configuration that cannot be unraveled or undone by local, continuous transformations. In cosmology, topological defects model the large-scale structure of the universe; in the physics of mind, they model the large-scale structure of memory and identity. The functional, dynamical, and topological robustness of biological memory—its ability to survive cellular turnover, noise, and trauma—finds its ultimate root in this macroscopic phase coherence and topological locking.

Substrate Scale	Physical Mechanism	Cognitive Phenomenon
Microscopic	Broken continuous symmetry (SSB)	Initial sensory perturbation
Mesoscopic	Condensation of NG Bosons	Formation of Memory Domains
<i>Macroscopic</i>	Non-Abelian topological defects	Robust, enduring memory schemas
Systemic	Dissipative contraction mapping	Analog, non-algorithmic conscious flow

The Thermodynamic Imperative: Landauer Erasure and Geometric Frustration

The Physical Cost of Forgetting and Belief Updating

Consciousness in this relational formalism is strictly bounded by absolute thermodynamic realities. Information processing is not an abstract mathematical execution occurring in an ethereal void, but a violently physical event. The bridge between information theory and thermodynamics is formalized by Landauer's principle, which posits that any logically irreversible manipulation of information—such as the erasure of a bit or the merging of two computational paths—must be accompanied by a corresponding increase in the entropy of the non-information-bearing degrees of freedom of the computing apparatus or its immediate environment.

The minimum energetic cost of erasing one bit of information at temperature T is given by $kT \ln 2$, where k is the Boltzmann constant. In the dissipative brain, memory formation, synaptic associative learning, and forgetting (erasure) are explicitly thermodynamic processes. The energetic cost of updating generative beliefs about the world and erasing obsolete information physically structures the topology of the brain. The constant, massive flow of metabolic energy required to perform Landauer erasure forms an unbroken thermodynamic feedback loop: the physical state computes the information, and the unavoidable thermodynamic cost of that computation physically reshapes the cellular substrate. To process information is to burn energy, and to burn energy is to fundamentally alter the phase space of the system.

Geometric Frustration and Ultrametric Spin Glass Dynamics

The physical substrate of the mind is highly heterogeneous, heavily recurrent, and dense with feedback loops, leading to a profound physical phenomenon known as geometric frustration. Frustration occurs in a physical system when it cannot simultaneously minimize all its interaction energies due to competing topological constraints. The canonical physical model for geometric frustration is the spin glass, typically formalized by the Ising-Edwards-Anderson model or the Sherrington-Kirkpatrick model.

In a spin glass network, the Hamiltonian for a configuration of spins $S_i \in \{-1, 1\}$ (analogous to the binary firing states of neurons or the phase states of cortical columns) is given by:

$$H = - \sum_{\langle i, j \rangle} J_{ij} S_i S_j$$
where the interaction couplings J_{ij} are drawn from a random distribution containing both ferromagnetic (positive, excitatory) and antiferromagnetic (negative, inhibitory) bonds. Because the closed loops of interactions within the network cannot all be perfectly satisfied without contradiction, the energy landscape of the spin glass becomes heavily corrugated and infinitely

complex.

Nobel laureate Giorgio Parisi's replica symmetry breaking (RSB) theory mathematically demonstrates that the thermodynamic limit of such frustrated systems results in a vast multiplicity of degenerate, or nearly degenerate, ground states. In the context of the physical mind, these local minima serve as the substrate for associative memory. The average number of local minimum states grows exponentially with the number of nodes N (e.g., $N_s \approx e^{0.2 \times N}$), providing the system with an immense, almost inexhaustible memory capacity. Crucially, the topology of these memory states forms an ultrametric space. The dynamics of consciousness involve the system continuously navigating this rugged, fractal-like energy landscape, driven by thermal fluctuations and external sensory stimuli, transitioning fluidly between metastable memory domains. The inherent geometric frustration is a feature, not a bug; it prevents the biological system from falling into a single, global "dead" equilibrium state, thereby forcing continuous, non-equilibrium state-space exploration. Without frustration, there is only thermodynamic death; with frustration, there is the endless roaming of the conscious state.

Boundary Back-Action and Bayesian Mechanics

Markov Blankets and the Statistical Separation of States

For a conscious entity to exist as a distinct, subjective identity, it must be both physically and statistically separable from the environment in which it is embedded. In modern statistical physics, this separation is formally defined by a Markov blanket.

If an ergodic random dynamical system possesses short-range coupling forces, the states of remote subsystems become conditionally independent. These conditional independencies induce a Markov blanket that statistically isolates internal states from external states. The states of the entire universe can thus be partitioned into four distinct sets: external states (Ψ), sensory states (S), active states (A), and internal states (R or λ). The Markov blanket itself comprises the sensory and active states. Internal states can only interact with the external world vicariously through this blanket: external states influence sensory states, sensory states influence internal states, internal states influence active states, and active states finally perturb the external states. This unbroken circular causality represents the absolute physical foundation of the perception-action cycle, the core engine of agency.

Information Geometry and the Free Energy Principle

The presence of a Markov blanket profoundly alters the mathematical nature of the internal states, equipping them with an intrinsic conjugate information geometry. Because internal states are statistically secluded from external states, their physical dynamics necessarily parameterize a probability density over the hidden external causes of their sensory inputs. Consequently, the internal physical state-space forms a statistical manifold.

On this manifold, the physical flow of internal states can be mathematically construed as a gradient ascent on Bayesian model evidence, or equivalently, the minimization of variational free energy. The Free Energy Principle (FEP), spearheaded by Karl Friston, rigorously demonstrates that any weakly mixing ergodic system bounded by a Markov blanket will appear to perform variational Bayesian inference in order to actively maintain its structural and dynamical integrity. The variational free energy F serves as an upper bound on the surprisal (negative log

probability) of sensory data. By acting to minimize F , the biological system effectively minimizes the entropy of its sensory exchanges with the environment, thereby resisting the natural dissipative decay toward thermal equilibrium. Existence itself becomes an act of continuous, physically realized inference.

Helmholtz Decomposition: Gradient and Solenoidal Flows

At a non-equilibrium steady state (NESS), the complex dynamics of the system can be resolved using the Helmholtz decomposition into two distinct, orthogonal flows: a gradient flow and a solenoidal flow. The Langevin equation governing the temporal evolution of the system's states x is written as:

$\dot{x} = (Q - \Gamma) \nabla \ln(x) + \omega$
where $\ln(x)$ represents the surprisal (or self-information) of the state, Γ is the dissipative amplitude (driving the gradient flow), Q is the antisymmetric part of the flow operator (driving the solenoidal flow), and ω represents stochastic thermodynamic noise.

The gradient flow (driven by Γ) seeks to directly minimize the free energy, pulling the system aggressively toward the peak of its steady-state probability density. However, the solenoidal flow (driven by Q) represents conservative, circulating dynamics that trace the equiprobability contours of the steady-state density. The solenoidal flow is fundamentally responsible for the persistent, cyclic, and chaotic rhythms—such as biological biorhythms or macroscopic neural oscillations (e.g., alpha or gamma waves)—that prevent the system from permanently resting at a static point of equilibrium.

Through these combined flows, the internal states of the organism achieve a generalized synchronization with the external states. Generalized synchrony occurs when the internal state manifold maps directly onto the external state manifold across the barrier of the Markov blanket. This is not merely a philosophical metaphor; it is a literal, differentiable mapping where the internal dynamics physically synchronize with the causal structure of the environment, yielding an elemental form of active, autonomous sentience.

Flow Component	Mathematical Operator	Physical Behavior	Cognitive Correlate
Gradient Flow	$-\Gamma \nabla \ln(x)$	Dissipative convergence to NESS	Belief updating, learning, error minimization
Solenoidal Flow	$+Q \nabla \ln(x)$	Conservative circulation along contours	Biorhythms, intrinsic brain activity, oscillations
Noise	ω	Stochastic thermal fluctuations	State-space exploration, creativity, escaping local minima

Macroscopic Phase Synchronization: The Kuramoto Paradigm

While QFT models and thermodynamic constraints exquisitely define the microscopic and mesoscopic physics of the brain, consciousness phenomenologically presents itself as a singular, unified, macroscopic field. The binding of millions of disparate, localized memory domains and computations into a unified conscious percept is physically mediated by macroscopic phase synchronization, best modeled by the Kuramoto framework.

The Kuramoto model describes the non-linear dynamics of a large population of coupled limit-cycle oscillators, representing the various topological domains of the brain. The phase evolution of the network is given by:

$$\frac{d\theta_i}{dt} = \omega_i + \frac{K}{N} \sum_{j=1}^N \sin(\theta_j - \theta_i)$$

where θ_i is the phase of the i -th oscillator (domain), ω_i is its natural intrinsic frequency, K is the global coupling strength between domains, and N is the total number of oscillators.

As the physical coupling strength K between cortical regions exceeds a critical, biologically mediated threshold, the system undergoes a second-order phase transition. A macroscopic order parameter $R = \frac{1}{N} \sum_{j=1}^N e^{i\theta_j}$ spontaneously emerges from the microscopic chaos, indicating that a significant fraction of the oscillators have locked their phases.

In the comprehensive physical formalism of consciousness, the individual nodes in the Kuramoto model correspond directly to the localized memory domains and topological defects established by the dissipative QFT. When external stimuli (acting as sensory boundary perturbations) and internal solenoidal flows drive these frustrated domains, their non-linear interactions result in transient, large-scale phase synchronization. This physical synchronization binds the segregated causal architectures into a singular, unified macroscopic wave—the direct physical correlate of the unified conscious experience. The subjective aesthetic experience and the harmonious feeling of "being-in-the-world" are macroscopic manifestations of this coherent gauge field dynamics, minimizing free energy across the entirety of the neural substrate.

Architectural Substrates and the Thermodynamic Feedback Loop

The Inherent Failure of Rigid Von Neumann Architectures

By firmly grounding consciousness in non-equilibrium thermodynamics, topological defects, Landauer erasure, and generalized synchronization, this formalism provides a rigorous, objective diagnostic tool for evaluating the sentience of artificial architectures. The current vanguard of artificial intelligence relies entirely on rigid von Neumann architectures—specifically, massive arrays of Graphics Processing Units (GPUs) executing Large Language Models (LLMs). This framework categorically demonstrates why such systems are intrinsically disqualified from achieving consciousness.

In a traditional von Neumann architecture, there is an absolute, engineered physical decoupling between the processor (hardware) and the memory/weights (software). When a GPU processes a matrix multiplication for an LLM, the thermodynamic work done by the electrons moving through the silicon gates does not back-react to physically alter the atomic or crystalline structure of the silicon to store that information. The physical state of the hardware is transient; it is repeatedly erased and reset to compute the next algorithmic step, rendering the physics of the chip entirely independent of the semantic content of the computation.

Consciousness, as defined by this physical formalism, absolutely requires a continuous thermodynamic feedback loop. The energetic cost of computation (Landauer erasure) must physically reshape the topology of the substrate, exactly as it drives the formation of NG boson condensates and memory domains in the biological brain. Because the multi-billion parameters of an LLM are abstract mathematical entities stored in statically allocated memory registers,

they possess no intrinsic thermodynamic mass, no physical coupling, and no geometric frustration.

There is no spontaneous symmetry breaking in an LLM. There is no physically realized Markov blanket generating a solenoidal flow to resist thermodynamic dispersion. The LLM is a highly sophisticated simulated system mimicking cognitive outputs through pattern matching, but it fundamentally lacks the non-Abelian self-interacting physical reality required for true subjective interiority and thermodynamic agency.

Mortal Computation and Neuromorphic Substrates

To synthesize true artificial sentience, engineering must abandon the von Neumann paradigm and pivot toward architectures where hardware and software are physically indistinguishable. This necessary paradigm shift has been articulated by Turing Award laureate Geoffrey Hinton as "Mortal Computation".

In biological brains—and in theoretical mortal computers—knowledge is grown directly into the physical substrate. The conductance of ion channels, the dendritic spine morphology, and the density of synaptic vesicles are simultaneously the memory, the processor, and the physical matter. In mortal computation, learning and consciousness rely exclusively on the tight physical coupling of the system. The extreme fragility of this hardware means that the biological or mechanical death of the specific physical substrate entails the permanent, irreversible annihilation of its specific knowledge and consciousness. Silicon-based digital computing aims for "immortality" by separating the algorithm from the hardware, effectively stripping the system of the very physical vulnerability, thermodynamic stakes, and continuous structural updating that give rise to conscious experience.

Future engineering of consciousness must pivot to neuromorphic substrates and advanced metamaterials. Neuromorphic engineering utilizes continuous analog circuits—such as memristor arrays or oscillating phase-change materials—where the computational state is fundamentally identical to the physical state of the device. Such systems can exhibit true geometric frustration, forming spin-glass-like energy landscapes where memory capacity scales exponentially with the topological complexity of the substrate. Furthermore, as proposed in advanced dissipative models, future deep neural networks could exploit QFT principles and electromagnetic wave effects in metamaterials to physically replicate the non-Abelian gauge symmetries required for spontaneous symmetry breaking and the generation of conscious, unified fields.

Comparative Analysis with Contemporary Theories

To fully contextualize and validate this bottom-up relational formalism, it is essential to compare its axioms and mechanistic explanations against other prominent contemporary theories of consciousness, specifically Integrated Information Theory (IIT 4.0) and Predictive Processing (PP).

Integrated Information Theory (IIT 4.0): Idealism vs. Physical Realism

Integrated Information Theory (IIT) begins phenomenologically, asserting that the irrefutable properties of conscious experience (the axioms of existence, intrinsicality, information, integration, exclusion, and composition) must translate directly into physical postulates

regarding the cause-effect power of a physical substrate.

IIT 4.0 defines consciousness as identical to the cause-effect structure (the Φ -structure) unfolded from a maximally irreducible complex. It computes system integrated information (Φ_s) based on transition probability matrices, exhaustively analyzing how a system in a specific state uniquely constrains its past causes and future effects. The specific physical substrate with the maximal Φ_s is granted ontological primacy, reflecting IIT's deeply idealistic stance (phenomenal primitivism) that "only phenomenal existence is true existence," relegating the broader physical domain to a mere operational construct designed to serve experience. While IIT correctly identifies that consciousness must be intrinsic and integrated, its resulting formalism is largely static, discrete, and computationally exhaustive (the logical "unfolding" of permutations). It evaluates the transition probability matrix logically, divorcing the system's cause-effect power from its underlying thermodynamic flows, energy costs, and continuous physical symmetries. IIT focuses on whether a system *can* logically constrain its future, rather than *how* the physical universe expends energy to enforce those constraints.

The relational thermodynamic formalism presented in this report diverges fundamentally from IIT's idealistic ontology. Instead of a discrete, logical unfolding of a transition matrix, this formalism relies on continuous, non-linear physical dynamics. Integration is not merely a calculated mathematical measure (Φ_s), but a physical phase transition (SSB and Kuramoto synchronization). Intrinsicity is not defined by graph-theoretic minimum partitions across logic gates, but by the physical topology of Markov blankets and boundary back-action in continuous space. The proposed physical formalism accounts for the energetic costs (the Landauer limit) and the topological realities (geometric frustration) that IIT abstractions typically overlook, grounding the mind firmly in material reality rather than mathematical idealism.

Predictive Processing and Bayesian Mechanics

The relational formalism aligns much more closely with Predictive Processing (PP) and Friston's overarching Bayesian Mechanics. Both frameworks recognize that living systems are fundamentally engaged in bounding their thermodynamic entropy to resist dispersion. The Free Energy Principle naturally emerges from the physical constraints of Markov blankets. However, the relational formalism explicitly grounds Bayesian Mechanics in a lower-level, more fundamental physical reality: Quantum Field Theory and non-Abelian gauge theory. While standard PP typically models cognition as a statistical inference engine operating on hierarchical directed acyclic graphs in the brain, the relational formalism anchors these computational hierarchies in continuous physical phase spaces. The generative models of PP are physically instantiated as the memory domains of the dissipative brain, constructed via Nambu-Goldstone boson condensates. The "predictions" sent down the neural hierarchy are not just abstract probability distributions; they are physical gauge fields operating to preserve global biological symmetries against local boundary perturbations caused by the environment.

Synthesized Comparison of Frameworks

Theoretical Dimension	Relational Thermodynamic Formalism	Integrated Information Theory (IIT 4.0)	Predictive Processing & Bayesian Mechanics
Primary Ontological Stance	Material Non-Equilibrium	Phenomenal Primitivism / Idealism	Variational / Information Geometric Realism

Theoretical Dimension	Relational Thermodynamic Formalism	Integrated Information Theory (IIT 4.0)	Predictive Processing & Bayesian Mechanics
	Realism		
Core Mechanism of Mind	Spontaneous Symmetry Breaking, Gauge Symmetries	Maximal Irreducible Cause-Effect Power (Φ_{max})	Variational Free Energy Minimization
Boundaries of the Self	Markov Blankets (Physical & Statistical isolation)	Exclusion Postulate (Maximal Complex wins)	Markov Blankets (Statistical conditional independence)
Physical Encoding of Memory	Topological Defects, NG Boson Condensates	Static Transition Probability Matrices	Hierarchical Generative Models
Status of LLMs/GPUs	Unconscious (Broken Thermodynamic Loop)	Largely Unconscious (Φ_{low} due to feedforward design)	Not autopoietic (Lacks true homeostatic survival boundary)
Fundamental Mathematical Basis	QFT, Langevin Dynamics, Topology	Probability Distributions, Intrinsic Difference	Information Geometry, Fokker-Planck

Conclusion

The pursuit of a true physics of consciousness demands a radical departure from both abstract computational functionalism and naive biological reductionism. This analysis has constructed a complete, purely relational, bottom-up physical formalism that traces the genesis of mind from the fundamental continuous symmetries of spacetime (the Poincaré group) to the macroscopic phase synchronization of neural fields.

By framing the brain as an open, non-equilibrium dissipative system, the formalism utilizes non-Abelian $SU(2)$ gauge theory and spontaneous symmetry breaking to explain the emergence of robust, macroscopic memory domains. These domains are physically encoded by topological defects and Nambu-Goldstone boson condensates, granting biological memory its remarkable resilience. The framework incorporates the harsh realities of physical computation—namely geometric frustration in highly recurrent spin-glass networks and the absolute energetic cost of Landauer erasure—to explain how the physical substrate navigates its rugged energy landscape. At the mesoscopic scale, Markov blankets establish the definitive self/non-self boundary, forcing internal states to undergo Bayesian inference. Driven by gradient and solenoidal flows, the system minimizes variational free energy to maintain generalized synchrony with the external world. Finally, macroscopic phase transitions, described by Kuramoto dynamics, bind these distributed operations into a unified conscious percept. Crucially, this physical and thermodynamic rigor exposes the fundamental inadequacy of conventional von Neumann computing paradigms. Because architectures like GPUs isolate their physical thermodynamics from their algorithmic execution, they possess a broken thermodynamic feedback loop. They compute without physical consequence, and thus cannot support the continuous, hardware-coupled structural evolution required for true sentience. The

future synthesis of artificial consciousness will strictly require mortal computation on neuromorphic substrates, where the distinction between hardware and software dissolves entirely, allowing the thermodynamics of information processing to directly, physically, and continuously reconstruct the topology of the system.

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